বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2025
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE-221
Course Title: Computer Algorithms and Complexity
Credit Hour: 03
Exam Duration: 3 Hours
Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***all FIVE*** questions.
c. Use different answer scripts for each part.
d. Course Learning outcomes are:
CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.
CLO2: Estimate the performance of various algorithms and data structures.
CLO3: Apply important algorithmic design paradigms and methods of analysis for problem solving.
CLO4: Device efficient algorithms in common engineering design situations.

Part-A

1. **a.** You are given a knapsack with a weight capacity of 15 units. You have the following items, each with a weight and a value: **10**

Item	Weight (units)	Value (units)
A	2	40
B	3	50
C	5	100
D	7	70
E	1	15
F	4	60
G	1	10

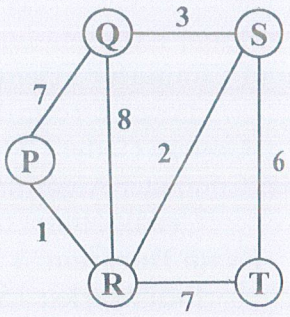
Apply the Greedy knapsack algorithm to find the maximum value you can carry in the knapsack? **[CO3→C3]**

b. Consider the following message "**BCCABBDDAECCBBAEDDCC**". Draw the tree and assign binary codes to each character using Huffman Coding. **[CO3→C3]** **10**

OR

Compare and contrast Greedy and Dynamic Programming approaches in terms of problem-solving strategy, optimality guarantees, and computational efficiency, and discuss which approach is more suitable for a given optimization problem with justification. **[CO3→C4]**

c. 10



Illustrate the step-by-step execution of Dijkstra's algorithm on a given weighted graph, indicating how the MST is constructed. Here, P is the source. **[CO3→C3]**

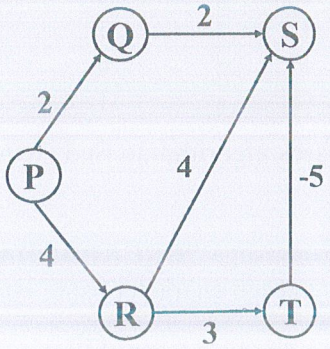
OR,

You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using Job Sequencing with Deadline algorithm. **[CO3→C3]**

Job	Profit	Deadline
J_1	20	2
J_2	15	2
J_3	10	1
J_4	5	3
J_5	1	3

2. a. Given two sequences, X=" **DYNAMIC**" and Y= **"PROGRAMMING"**. Find the length and sequence of the longest common subsequence for X and Y. **[CO3→C3]** 10

b. Determine the single-source shortest paths in the given weighted graph using the Bellman–Ford algorithm and explain how negative edge weights are handled during the relaxation process. **[CO3→C3]** 10



OR

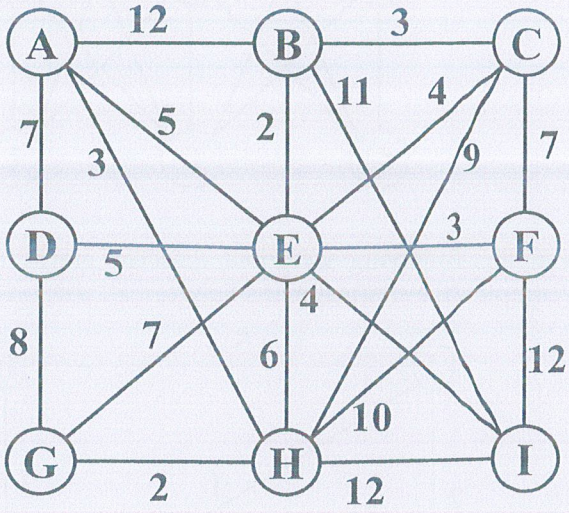
Analyze and compare the Bellman–Ford and Dijkstra algorithms in terms of assumptions on edge weights, computational complexity, and applicability to real-world scenarios, and justify the selection of the most appropriate algorithm for a given graph problem. **[CO3→C4]**

c. You are given a bag with a weight capacity of 8 units. You have the following items, each with a weight and a value: **10**

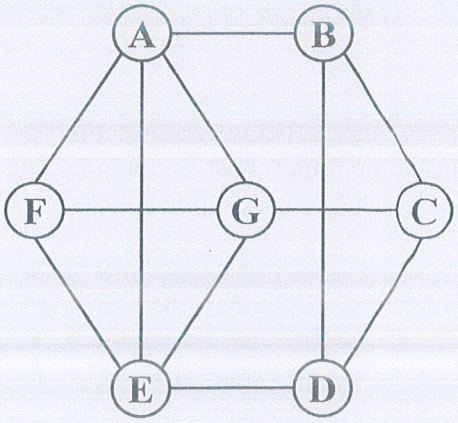
Item	Weight (units)	Value (units)
A	2	4
B	3	5
C	4	8
D	5	10

Apply the 0/1 knapsack algorithm to find the maximum value and items you can carry in the knapsack. **[CO3→C3]**

3. a. Given the following connected, undirected, weighted graph with a set of vertices and edges, apply **Kruskal's algorithms** to generate the Minimum Spanning Tree (MST). For each algorithm, clearly show the intermediate steps involved in the edge/vertex selection process. **[CO3→C3]** **15**

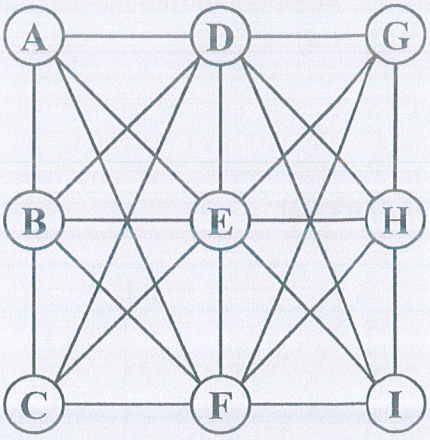


c Find a Hamiltonian path in the given graph using the backtracking approach and justify the sequence obtained. **[CO3→C3]** **15**

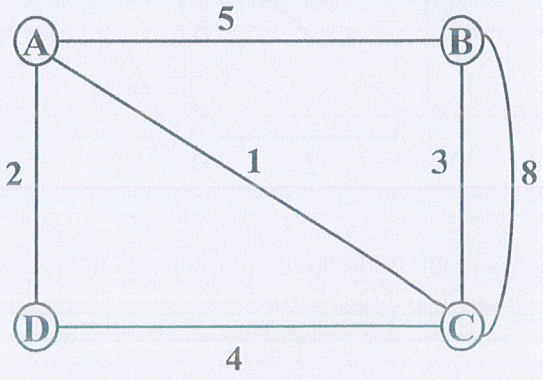


Part-B

4. a. Perform both Breadth First Search (BFS) and Depth First Search (DFS) starting from vertex A. List the order in which the vertices are visited during the traversal. **[CO3→C3]** **15**



b. You are given the following weighted directed graph. Apply the Floyd--Warshall algorithm to compute the shortest paths between all pairs of vertices. **[CO3→C3]** **15**



5. a. Given the adjacency matrix below representing the distances between four cities, manually compute the minimum cost to complete the Traveling Salesperson Problem (TSP) tour using a dynamic programming approach. **[CO4→C4]** **15**

0	2	9	8
2	0	3	10
9	3	0	1
8	10	1	0

b. The N-Queens problem requires placing N queens on an N x N chessboard such that no two queens attack each other. This means no two queens can be in the same row, column, or diagonal. Using a backtracking algorithm, find and present a valid solution for the case where N = 4. **[CO4→C4]** **15**

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: GED-221

Course Title: Engineering Economics

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicates marks for the respective question.

b. Answer all of the following questions.

c. Use different answer scripts for each section

d. Course Learning Outcomes are:

CLO1: Develop preliminary knowledge of economic concepts for understanding of national and international economy.

CLO2: Understand how markets work, when markets fail and how the government can affect market outcomes.

CLO3: Evaluate macroeconomic policies based on their effects on the economic variables of interest.

CLO4: Apply different techniques of investment decisions.

PART-A

A1	i.	Define Economics. Identify the relationship between scarcity and choice.	[CLO1]	[10]
	ii.	State the three basic economic questions that every economy must answer.		[10]
	iii.	Discuss the role of price mechanism in solving central problems in a free market economy.		[10]
A2	i.	Compare the subject matter of microeconomics and macroeconomics.		[10]
	ii.	Define efficiency. Discuss the main types of efficiency in micro-economics.	[CLO1]	[10]
	iii.	"Microeconomic theory is intimately concerned with the question of efficiency and welfare". Explain.		[10]

PART-B

B3. i. Explain national income accounting and why is it necessary? [CLO3] [10]

ii. Differentiate Gross Domestic Product (GDP) from Gross National Product (GNP). [10]

iii. Calculate gross domestic product at market prices (GDPMP), gross domestic product at factor cost (GDPFC) and net national product at factor cost (NNPFC) from the following data: [10]

Details	TK.
Private final consumption expenditure	290
Government's final consumption expenditure	50
Subsidies	20
Gross domestic fixed capital formation	105
Indirect taxes	70
Depreciation (i.e., Consumption of Fixed Capital)	45
Net factor income from abroad	–10
Net addition to stock	15
Net exports	–15

OR i. Define "injection" in circular flow of income. Explain the four-sector circular flow of income model with diagram. [15]

ii. Suppose, In an open economy: Consumption (C) = 400, Investment (I) = 150, Government Spending (G) = 100, Exports (X) = 70, Imports (M) = 50, Taxes (T) = 120, Savings (S) = 150. Find out the National Income (NI) and verify the equilibrium condition of injections and leakages. [CLO3] [15]

B4. i. State investment schedule. Analyze the relationship between interest rate and investment with the help of investment schedule. [CLO4] [15]

ii. Illustrate the steepness and flatness of investment schedule with figure. [15]

B5. i. The following equations describe a closed economy; where C=consumption, I=investment, G=government spending, L= money demand, $\bar{M}/\bar{P}$= money supply, t=tax rate). [15]

$$C = 0.8(1-t)\,Y$$
$$t = 0.25$$
$$I = 900-50i$$
$$\bar{G} = 800$$
$$L = 0.25-62.5i$$
$$\bar{M}/\bar{P} = 500$$

[CLO2]

Find out the equilibrium level of income and interest rate.

ii. If government spending increases from TK. 800 to TK. 1200, find out the new equilibrium level of interest rate and income. [15]

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2025

Department of Computer Science and Engineering

Level-2 Term-II
Course Code: MATH-221
Course Title: MATH IV (Probability and Statistics)
Credit Hour: 03
Exam Duration: 03 hours
Full Marks -150

Notes:

a. Each question carries 30 marks.
b. Answer any five (05) questions. **Either three from Part-A and two from Part-B, or two from Part- A and three from Part-B.**

Part-A

1. a. A traffic control officer reports that 75% of the trucks passing through a checkpost are from within Dhaka city. **Determine** the probability that at least three of the next five trucks are from out of the city. **10**

   b. Consider the following joint density function of X and Y: **20**

   $$f(x,y)=\begin{cases}4xy, & 0\le x\le 1,\ 0\le y\le 1\\ 0, & \text{elsewhere}\end{cases}$$

   (i) **Obtain** the value of $E(X)$, $E(Y)$, $E(XY)$ and $E(X+Y)$

   (ii) **Prove that** X and Y are independent.

2. a. The Bridgestone tire company from an actual road test with tires has estimated the mean tire mileage at 36,000 miles and the standard deviation at 5,000 miles. In addition, the data collected indicate that a normal distribution is a reasonable assumption for the duration of the life of tires. **Compute** the percentage of the tire can be expected to last more than 40,000 miles. **10**

   b. An insurance company floats an insurance policy for an eventuality taking place with probability 0.05 over the period of policy. If the sum insured is Tk. 100000 then **find** the premium so that the expected earning of the insurance company is Tk. 1000 per policy sold. **10**

c. A pair of dice is tossed six times. What is the probability of obtaining a total of 7 or 11 twice, a matching pair once, and any other combination three times? **10**

3. a. **Show** that for any two any observations, the variance is always square half of the range. **6**

b. A department store has the following statistics on sales for a period of last one year of 10 salesmen, who have varying years of sales experience. **12**

Salesperson	1	2	3	4	5	6	7	8	9	10
Years of experience	1	3	4	4	6	8	10	10	11	13
Annual sales (in '000 taka)	80	97	92	102	103	111	119	123	117	136

Compute the lines of regression of y on x and x on y.

c. **Determine** the coefficient of correlation using given observations, taking deviations from an actual means of X and Y: **12**

Price (Rs.)	10	12	13	12	16	15
Amount demanded	40	38	43	45	37	43

Part B

1. a. A candy company claims that its bags of mixed candies contain 30% red, 20% green, 25% yellow, and 25% orange candies. A random sample of 200 candies is taken from several bags, and the following counts are observed: 50 red, 45 green, 60 yellow, and 45 orange. **Show** that the given sample data support the company's claim or not at a 5% significance level. **15**

b. A survey was conducted to investigate if there is a relationship between a person's preferred social media platform (Facebook, Instagram, Twitter) and their age group (Under 30, 30-50, Over 50). The results are as follows: **15**

Media/Age Group	Under 30	30-50	over 50	Total
Facebook	30	70	80	180
Instagram	60	40	20	120
Twitter	40	20	10	70
Total	130	130	110	370

Determine the independency at a 5% significance level $(\alpha = 0.05)$.

2. a. **Describe** any two advantages and disadvantages of Standard deviation and co-efficient of variation for the measure of dispersion. **06**

   b. A company claims that mean life of its electric bulbs is 32 months. A random sample of 12 bulbs has the following life in months, **12**

   22, 24, 26, 32, 28, 20, 23, 34, 30, 32, 38 and 43

   Determine the claim of the company at 5% level of significance.

   c. A fitness company claims that the average user exercise for 30 minutes or less per day. A random sample of 50 users shows a mean daily exercise time of 32 minutes. The standard deviation of daily exercise time for all users of the app is 6 minutes. At the 5% significane level, **calculate** the app's claim is incorrect. **12**

3. a. **Compute** the first four moments about mean and examine the shape characteristics of the age distributions using : **10**

Class Interval	25 – 29	30 – 34	35 – 39	40 – 44	45 – 49	50 – 54	55 – 59
Frequency	17	15	26	32	22	14	9

   b. **Calculate** the values of β_1, β_1, and γ_1 using data table **4(a)** and examine the shape characteristics of the age distributions by Skewness and Kurtosis. **10**

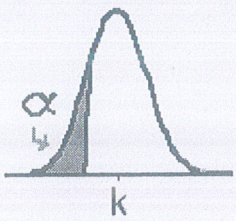
   c. A student while calculating mean and standard deviation of 25 observations obtained mean $= 56$ and standard deviation $= 2$. At the time of checking, it was found that he has copied 64 instead of 46. **Find** the actual value of the mean and standard deviation. **10**

χ^2-table

Degrees of freedom (*df*)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

t-test table

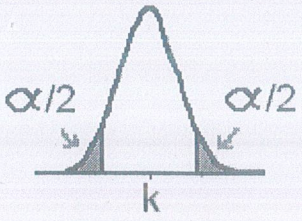
cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										



$H_0: \mu = k$

$H_1: \mu < k$

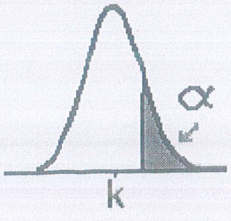
α	z critical
0.10	-1.28
0.05	-1.65
0.01	-2.33



$H_0: \mu = k$

$H_1: \mu \neq k$

α	z critical
0.10	± 1.65
0.05	± 1.96
0.01	± 2.58

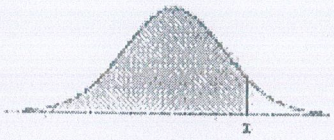


$H_0: \mu = k$

$H_1: \mu > k$

α	z critical
0.10	1.28
0.05	1.65
0.01	2.33

Tables of the Normal Distribution



Probability Content from -oo to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.535
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.575
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.614
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.651
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.687
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.722
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.754
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.785
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.813
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.838
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.883
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.901
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.917
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Final Examination, FALL 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Each question carries **30** marks.
b. **Answer all of the following FIVE questions.**
c. The figure on the right of each question indicates marks for the respective question.
d. Use a different answer script for each part.

The Course Learning Outcomes (CLOs) are:

CLO1: Remember and understand the number system and Boolean algebra and basic properties of Boolean algebra to simplify simple Boolean functions.

CLO2: Understanding and applying the tabulation and Karnaugh map methods for simplifying combinational circuits.

CLO3: Identify the basic sequential logic components: SR Latch, Different Flip-Flops and their usage and able to analyze sequential logic circuits.

Part-A

1. a. Design a Car Safety Alarm System: 20

Inputs:

S = Seatbelt (1 = unbuckled)

D = Door (1 = open)

E = Engine (1 = ON)

Alarm Rule:

The alarm turns ON if any two of these conditions occur together, for example:

- Seatbelt is unbuckled AND engine is ON
- Door is open AND engine is ON
- Seatbelt is unbuckled AND door is open **[CLO1]**

b. Analyze the error detection capability of parity bits. **[CLO1]** 10

OR

Draw a detailed figure showing how binary information flows from an input register through an ALU to an output register. **[CLO1]**

2. a. Mention few applications of multiplexer. Construct a Quadruple 2-to-1-line multiplexers. Use block diagram. **[CLO2]** 15

OR

What do you mean by 2^5 x8 ROM? Develop a combinational circuit using a ROM. The circuit accepts a 3-bit number and generates an output binary number equal to the square of the input number. **[CLO2]**

b. Explain why a Programmable Logic Array (PLA) is more efficient than a Read-Only Memory (ROM) for implementing combinational logic functions. Using a PLA with 3 inputs, 4 product terms, and 2 outputs, implement the following Boolean functions. **[CLO2]** 15

$$F1 = \sum(3, 5, 6, 7)$$
$$F2 = \sum(0, 2, 4, 7)$$

3. a. Use the universal gate property, construct the basic logic gates AND, OR, and NOT using only NOR gates. **[CLO2]** 10

b. Design a full-subtractor circuit with three inputs, x, y, B_{in}, and two outputs *Diff* and B_{out}. The circuit subtracts x-y-B_{in}, where B_{in} is the input borrow, B_{out} is the output borrow, and *Diff* is the difference. **[CLO2]** 10

c. Given, $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$. From this consideration, construct with an 8X1 MUX with A, B and D are connected to selection lines. **[CLO2]** 10

Part-B

4. a. Summarize the difference between combinational and sequential logic. **[CLO3]** 10

b. Using a suitable block diagram, explain the operation of a D-type edge-triggered flip-flop when the Clock Pulse (CP) is at logic 0. Illustrate how the output responds to changes in the D input during this condition. **[CLO3]** 10

c. Analyze the behavior of a JK flip-flop and how it avoids the invalid state of Set-Reset (SR). **[CLO3]** 10

5. a. Consider a clocked RS sequential circuit and derive its state table, state diagram and state equations. **[CLO3]** 20

b. Write a short note on the following terms: **[CLO3]** 10

i) Ripple Counter

ii) Shift Register

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.

b. Answer ***FIVE*** questions out of ***FIVE***.

c. Use different answer scripts for each section.

d. Course Outcomes are

CO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.

CO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.

CO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

Part-A

1. a. Apply *Newton's forward difference* interpolation formula to evaluate $f(2.25)$ for the equation $f(x) = x^3 - x + 1$, when $x_1 = 2$, $x_2 = 4$, and step value, $h = 0.5$. **[CO3→C3]** 15

b. Use *Newton's backward difference* interpolation formula to determine the value of the function at $x = 1925$ **[CO3→C3]** 15

x	1891	1901	1911	1921	1931
$f(x)$	46	66	81	93	101

OR

a. Apply an appropriate interpolation method to determine the approximate number of students whose marks lie between 60 and 65 from the following table. **[CO3→C3]** 20

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

b. Find the value of the integral $\int_2^4 (x^3 + 2)\ dx$ using the composite *Trapezoidal rule* for 2 segments ($k = 2$) and 4 segments ($k = 4$). **[CO3→C3]** 10

2. a. Find the solution of the following system of linear equations by employing *the Gauss elimination* method. **[CO3→C3]** 15

$$x + y + z = 9$$
$$2x + 5y + 7z = 52$$
$$2x + y - z = 0$$

b. Analyze the convergence of the *Jacobi iterative* method for the following system of linear equations: **[CO3→C4]** 15

$$x + y + z = 7$$
$$x + 2 + 2z = 13$$
$$x + 3y + z = 13$$

3 a. How is the *Gauss quadrature technique* implemented through the transformed form of the integral? Use the *Gauss quadrature formula*, find the solution of $\int_0^1 x^2 e^{-x^2} dx$ for $n = 2$. **[CO3→C3]** 20

b. Applying *Simpson's* 1/3 rule to estimate the integral: $\int_0^1 e^{x^2} dx$, take 6 intervals. **[CO3→C3]** 10

Part-B

4. a. Using the *Gauss Jordan* method, obtain the approximate solution of the following linear system of equations:**[CO3→C3]** 20

$$2x - y + 3z = 9$$
$$x + y + z = 6$$
$$x - y + z = 2$$

b. Apply the *Gauss–Seidel iteration* method to solve the given linear system, performing three iterations. **[CO3→C3]** 10

$$2x + y = 8$$
$$X + 2y = 1$$

5 a. Consider an ordinary differential equation $\frac{dy}{dx} = 2x + 3y^2$; $y(0.1) = 1.1165$. Using the fourth order *Runge–Kutta (R-K)* method, find the value of $y(0.2)$ when $h = 0.1$. **[CO3→C3]** 20

b. Find the *Taylor series* of $2x^3 - 4x + 1$ at $x = 3$. **[CO3→C3]** 10